Pierluigi Maria Teotini

MSc Mechanical Engineer · CFD, aerodynamics, composites

Mechanical engineer with an MSc in Advanced Automotive Engineering – Racing Car Design (MUNER, Dallara Academy), specialised in CFD, aerodynamics and composite materials. I built my CFD skills modelling a high-pressure H_2 injector at R&D CFD, in collaboration with Dumarey. Before that, I designed and manufactured aerodynamic components for Formula SAE. Years of top-level canoe polo gave me discipline, resilience and team spirit. I am now studying Python and data analysis, to enrich my technical background, and I am open to explore additive manufacturing and machine learning applications. I am looking for roles where I can keep learning and take on new technical challenges.

Experience

11/2025 – 05/2026

Thesis internship – CFD Analyst

R&D CFD SRL, Modena – in collaboration with Dumarey

- Steady and unsteady CFD approaches for modelling of a high-pressure H_2 injector in STAR-CCM+, with and without mesh motion (morphing, overset) and fluid-structure interaction (DFBI)
- Correlation of the simulations with experimental data

2018 – 2020

Aerodynamic team member

Formula SAE – Firenze Race Team

- Designed a multi-element front wing assembly in SolidWorks, validated with CFD
- Manufactured composite bodywork and aerodynamic components with prepreg and vacuum bagging

Selected projects

- **Sidepod inlet blanking, Formula 3 car.** Wind tunnel: cooling system optimization (-2.98% vs -3.0% target cross flow reduction) and 47% car balance tuning through FW polar
- **Composites laboratory.** DSC, DMA, tension, compression, shear and lap joint tests (T_g and mechanical characterization); Abaqus and Ply calculator design and simulation tools;
- **V6 engine vibration [Programmer and Team leader].** Six-degree-of-freedom modal model on elastic mounts in MATLAB, verified with ode45; FFT processing of transmitted forces

Education

2022 – 07/2026

MSc Advanced Automotive Engineering

Racing Car Design · MUNER, Modena – Dallara Academy · 108/110

Thesis: Analysis of CFD methodologies applied to a high-pressure H_2 injector in support to the design process - Supervisor: Stefano Fontanesi

2021 – 2022

Single courses, Aerospace Engineering

Politecnico di Torino

Applied aerodynamics, aeronautical structures, aeronautical propulsion

2016 – 2021

BSc Mechanical Engineering

University of Florence · 107/110

Thesis: gripper analysis for the capture of cooperative and non-cooperative space debris

Ongoing

Python programming and data analysis

Python Institute – independent study

Beyond engineering

2009 – Present

Canoe polo Athlete

ASD Canottieri Comunali Firenze

Canoeing since 2009; member of the Florence canoe polo team since 2012. Competed in the top Italian division. Won the Italian A1 Championship (2019). Won one U21 Italian Cup (2018).

2017-2021

Public events steward

Danza in Fiera, Pitti immagine

- Coordinator Fortezza da Basso's parking during "Danza in Fiera" events
- Security steward for "Pitti Immagine" events

Contact

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Florence, Italy
Open to relocation

linkedin.com/in/
pierluigi-maria-teotini

Portfolio: (pierr3x.github.io)

Languages

- Italian – native
- English – B2 (working proficiency)
- Spanish – B1

Skills

CFD and aerodynamics

STAR-CCM+ (advanced), Ansys Fluent and OpenFOAM (basic), turbulence modelling, mesh motion, fluid-structure interaction, wind tunnel testing, turbomachinery, propulsion

Structures and composites

Marc Mentat (advanced), Abaqus (intermediate), DSC, DMA, tension and compression tests, bonded joints, failure criteria, Ply calculator

Vibration and testing

impact hammer modal testing, FFT, modal analysis

CAD and design

SolidWorks (advanced), 3DEXPERIENCE (intermediate), Ansys Granta selector, GD&T, additive manufacturing

Programming

MATLAB (advanced), C (basic), Python and data analysis (learning), LaTeX

Soft skills

Teamwork, Problem solving, Leadership, Critical thinking, Stress management, Resilience, Ability to Plan and Organize

Certificates

Cambridge English, Level B2 (2016)

SolidWorks Associate, Mechanical Design (2022) | ID: C-X3FW9PJCGY